Senior AI Researcher Home Assignment

Dataset

Name: UTD-MHAD
Links: UTD-MHAD Website | Google Drive Folder

The dataset contains synchronized recordings of human actions captured simultaneously by multiple sensors.

For this assignment use:

RGB video and inertial sensors (accelerometer + gyroscope)

Choose a subset of classes and shorten sequences or sub-sample to keep runtime reasonable.

Objective

Build a system that performs **Human Activity Classification** using multimodal inputs. Additionally, the system must:

- Operate even when a modality is missing.
- Detect unreliable predictions.
- Identify unusual inputs.

Submission Guidelines

- Submit all code and documentation (py, ipynb, docx, pdf, md, etc.) in a single ZIP file.
- The code should be easy to run. Do not assume the reviewer has your specific local versions installed.
- Ensure results are reproducible.
- Include a HOW_TO_RUN.md file with clear execution instructions.

Part 1

Load the dataset and perform basic Exploratory Data Analysis (EDA).

Describe and visualize potential data issues and how they might affect a deployed system.

Part 2

Build two separate models (pre-trained models are allowed):

1. Sensor-only model
2. Video model

Report their performance and showcase a typical failure case for each.

Part 3

Build a model combining video and sensor inputs.

Explain your fusion design and evaluate:

- Does multimodal approach improve performance?
- In what specific scenarios does fusion help or fail?

Part 4

A. Simulate a scenario where one modality becomes unavailable at inference time. Evaluate your current model and propose a method to allow the system to continue operating.

B. If you knew during development that a modality might be missing at inference, how would you change your multi-modal design? Implement this re-design and compare its performance to the method in part A.

Part 5

Check for imbalance in your data set. If no imbalance is found, introduce it artificially.

Show: what fails / how you detect it / how you fix it (at least two techniques).

Part 6

Implement a confidence estimation mechanism and explain your chosen method (consider statistical guarantees).

Part 7

Introduce unseen actions and propose a method to detect out-of-distribution inputs.

Explain what the system should do when OOD is detected.

Part 8

Suppose you have a database of labeled past events.

- How can this data be used?
- When to use retrieval instead of fine-tuning? Provide advantages and risks for each approach.

Part 9

Assume the system is now in production. Describe:

- Key metrics to monitor.
- Methods for detecting data or concept drift.
- Logging strategies and retraining triggers.
- Provide a block diagram of your proposed system architecture.

Good Luck!

Rita WhatsApp 0544994207